

ABC COLLEGE OF ENGINEERING

CHOICE BASED CREDIT SYSTEM

Common to all B.E. / B.Tech. Full-Time Programmes

(For the students admitted to B.E. / B.Tech. Programme from the Academic year 2021- 2022 onwards)

1. DEFINITIONS AND NOMENCLATURE

In this Regulation, unless the context otherwise specifies:

- i. “Programme” means Degree Programme (i.e) B.E. / B.Tech. Degree Programme.
- ii. “Specialization” means discipline of B.E. / B.Tech. Degree Programme, like Mechanical Engineering, Bio Technology, etc.,
- iii. “Course” means a Theory or Practical subject that is normally studied in a semester, like Mathematics, Physics, Engineering Graphics, etc.,
- iv. “Controller of Examinations” means the Authority of the Institution who is responsible for all activities of the End Semester Examinations of the Institution.
- v. “Head of the Institution” means the Principal of the Institution.
- vi. “Head of the Department” means Head of the Department concerned.
- vii. “University” means ANNA UNIVERSITY, CHENNAI.

2. ADMISSION PROCEDURE

2.1. Admission to First Semester

- i. Students for admission to the first semester of the eight semester B.E./B.Tech. Degree Programme shall be required to have a pass in Higher Secondary Examination (Academic 10 + 2) Curriculum or its equivalent examinations with Mathematics, Physics and Chemistry/any other subjects accepted by the Directorate of Technical Education, Tamil Nadu and the affiliating University.

(OR)

- ii. Shall be required to have a pass in Higher Secondary Examination of Vocational Stream (Vocational groups in Engineering / Technology) as prescribed by the Directorate of Technical Education, Tamil Nadu and the affiliating University.

2.2. Lateral Entry Admission

- i. Candidates who possess Diploma in Engineering / Technology awarded by the State Board of Technical Education, Tamil Nadu or its equivalent are eligible to apply for Lateral entry 2 admission to the third semester of B.E. / B.Tech. Programme corresponding to the branch of study.

(OR)

- ii. The candidates who possess the Degree in Science (B.Sc.) (10+2+3 stream) with mathematics as a course at the B.Sc. level are eligible to apply for admission to the third semester of B.E. / B.Tech. Such candidates shall undergo two additional engineering courses in the third and fourth semesters as prescribed by the academic council of the institution.

3. UG PROGRAMMES OFFERED

A student may be offered admission to any one of the programme of study approved by the University. The recommended credit range for each programme is 165 –180,preferably around 170.

1. B.E. - Civil Engineering
2. B.E. - Computer Science and Engineering
3. B.E. - Electrical and Electronics Engineering
4. B.E. - Electronics and Communication Engineering
5. B.E. - Electronics and Instrumentation Engineering
6. B.E. - Mechanical Engineering
7. B.Tech. - Chemical Engineering
8. B.Tech. - Information Technology
9. B.Tech. - Bio Technology
10. B.Tech.- Artificial Intelligence and Data Science
11. B.Tech.- Artificial Intelligend and Machine Learning

4. STRUCTURE OF THE PROGRAMMES

4.1. Categorization of Courses

Every B.E. / B. Tech. Programme will have a curriculum with syllabi consisting of theory and practical courses that shall be categorized as follows:

- i. Humanities and Social Sciences including Management Courses (HSMC) include Technical English, Employability Skills, Engineering Ethics and Human Values, Communication skills and Management courses.
- ii. Basic Science Courses (BSC) include Mathematics, Physics, Chemistry, Biology, Environmental Sciences, etc.
- iii. Engineering Science Courses (ESC) include Engineering Practices, Engineering Graphics, 3 Basics of Electrical / Electronics / Mechanical / Computer Engineering, Instrumentation, etc.
- iv. Professional Core Courses (PCC) include the core courses relevant to the chosen specialization/branch.
- v. Professional Elective Courses (PEC) include the elective courses relevant to the chosen specialization/branch.
- vi. Open Elective Courses (OEC) include inter-disciplinary courses which are offered in other Engineering/Technology Programme of study
- vii. Employability Enhancement Courses (EEC) include Project Work, Internship, Career Development Skills, Creative and Innovative Project, Seminar, Professional Practices, Case Study and Industrial/Practical Training.
- viii. Mandatory courses (MAC) include Personality and Character development and the courses recommended by the regulatory bodies such as AICTE, UGC, etc.
- ix. Audit Courses (AC) expose the students to Unnat Bharath Abhiyan, Constitution of India, Essence of Indian Knowledge Traditional, Yoga, English for Research Paper Writing, Value education, Pedagogy Studies, Stress Management and Personality Development through Life Enlightenment Skills. Registration for any of these courses is optional to students.

4.1.1. Mandatory Personality and Character Development Programme

- All students shall enroll on admission in any one of the personality and character development programmes (NCC/NSS/NSO/YRC) and undergo training for about 80 hours and attend a camp of about seven days. The training shall include classes on hygiene, Women self esteem, self defense and health awareness and also training in first-aid.
- National Cadet Corps (NCC) will have about 20 parades.
- National Service Scheme (NSS) will have social service activities in and around the College / Institution. The activities will include hygiene and practical projects on recycling and reusing biodegradable and dry waste.
- National Sports Organization (NSO) will have sports, games, drills and physical exercises.
- Youth Red Cross (YRC) will have activities related to social services in and around College/Institutions. While the training activities will normally be during weekends, the camp will normally be during vacation period.

4.1.2. Mandatory Two Week Induction Programme

The students are expected to undergo a mandatory two week induction programme comprising of physical activity, creative arts, universal human values, proficiency modules, lectures by eminent people, visits to local areas and familiarization to department/branch & innovations immediately after admission as per UGC/AICTE guidelines.

4.2. Number of Courses per semester

The Curriculum of each semester shall normally have a blend of theory courses and theory cum practical courses not exceeding 6 and Practical courses not exceeding 3. In addition, Employability Enhancement Course(s) may also be included. Each course may have credits assigned as per clause 4.3. However, the total number of courses per semester shall not exceed 10 (including EEC). Pre-final semester may have 4 to 6 theory courses, Mini Projects and laboratory courses not exceeding 2. The final semester may have a blend of 2 theory courses and Project work.

4.3. Credit Assignment

Each course is assigned certain number of credits based on the following:

Contact period per week	Credits
One Period of Lecture(L) per week	1
One period of Tutorial (T) per week	1
2 Practical Periods (Laboratory / Seminar / Project Work / etc.)	1

The Contact Periods per week for laboratory can only be in multiples of 2. The length of the semesters shall be 15 to 18 weeks. Credit for a course shall vary from 1 to 4.

4.4. Internship

The students may undergo Internship for a period as specified in the Curriculum during the summer / winter vacation. The students may undergo Internship at a Research organization / University / industry (after due approval from the Heads of the Departments) continuously for the entire period.

If Internship is not prescribed in the curriculum, the student may undergo Internship optionally and the credits earned through the Internship shall be over and above the total credit requirement for the award of the degree.

Duration of Internship	Credits
2 Weeks	1
4 Weeks	2

4.5. Industrial Visit

Every student is required to go for at least one Industrial Visit every year, starting from the second year of the Programme. The Heads of Departments shall ensure that necessary arrangements are made in this regard.

4.6. Value Added Courses

Students may optionally undergo value added courses and the credits earned through the value added courses shall be over and above the total credit requirement prescribed in the curriculum for the award of the degree. The Head of the Department can constitute a committee, a team of senior faculty members (preferably three) to recommend the list of value added courses. One/two credit courses shall be offered by a department with the prior approval from the committee. The details of the syllabus, timetable and faculty may be sent to the controller of examinations after approval from the Head of the Department. Students can take a maximum of two 'one credit courses' or one 'two credit course'.

4.7. Online Courses

Students will be permitted to do online courses during 3rd to 8th semesters with final certification exams (NPTEL) to earn up to a maximum of six credits (which are provided with certificate after evaluation of the performance) with the prior approval from the Head of the Department. The Head of the Department can constitute a committee, a team of senior faculty members (preferably three) to recommend the list of online courses. On successful completion of each online course, the student has to submit the copy of the certificate to the Head of the Department. The committee recommends the credit and the grade that should be awarded to the student by appropriately mapping the score earned by the student. The results can be sent to the Controller of Examinations after the due approval by the Head of the Department. On successful completion of online courses adding to three credits, the student can obtain a waiver from doing either a Professional Elective or an Open Elective.

4.8. Medium of Instruction

The medium of instruction is English for all courses, examinations, seminar presentations and project/thesis/dissertations reports.

5. DURATION OF THE PROGRAMMES

5.1 The minimum and maximum periods for completion of the UG programmes are given below

Programme	Minimum No. of semesters	Maximum No. of semesters
B.E./ B.Tech.	8	14
B.E./ B.Tech. Lateral Entry	6	12

Each semester normally consists of 90 working days. In any contingent situation, the number of working days per semester shall not be less than 75 days. The Head of the Institution is given the discretionary powers to decide the number of working days in such contingencies.

The Head of the Institution shall ensure that every teacher imparts instruction as per the number of periods specified in the syllabus and that the teacher teaches the full content of the specified syllabus for the course being taught.

For regulations, the academic year has been divided into two semesters, the Odd semester normally spanning from July to November and the even semester from January to May.

The First semester of B.E. / B.Tech. Degree Programme normally spans from August to December and Second semester from January to May.

The total duration for completion of the programme reckoned from the commencement of the first semester to which the student was admitted shall not exceed the maximum duration specified in clause 5.1, irrespective of the period of break of study (vide clause 17) or prevention (vide clause 7.4), in order that the student may be eligible for the award of the degree (vide clause 14).

6. COURSE REGISTRATION

- i. Each student on regular admission has to register for all the courses prescribed in the first year of study.
- ii. A student has to earn the minimum number of total credits specified in the curriculum of the chosen programme of study in order to be eligible to obtain the degree. However, a student can earn more than the minimum number of credits, if he/she wish. In such cases, the minimum number of credits required for the completion of the programme alone would be considered for CGPA calculation.
- iii. The registration for the courses of the semesters III to VIII will commence 10 working days prior to the commencement of classes of the current semester. The student shall register for the courses with the guidance of his faculty mentor. The student may also drop courses (vide clause 6.1) within 5 working days of the commencement of the concerned semester and complete the registration process.
- iv. For an elective to be offered, the department shall specify the minimum number of students required.
- v. After registering for a course, a student should attend classes, satisfy attendance requirements (vide clause 7), earn continuous assessment and appear for the End Semester Examinations.

7. SYSTEM OF EXAMINATION AND ASSESSMENT PROCEDURE

Performance in each course of study shall be evaluated based on

- i. Continuous internal assessment throughout the semester
- ii. End Semester Examinations (ESE) at the end of the semester.

Each course, both theory / integrated and practical including project work shall be evaluated for a maximum of 100 marks. For all the End semester examinations, wherever necessary, the internal and external examiners shall be appointed by the Controller of Examinations.

Each course shall be evaluated for a maximum of 100 marks as detailed in Table

S.No	Category of course	Continuous Internal Assessments	End-Semester Examinations
1.	Theory Courses	40 Marks	60 Marks
2.	Theory Courses with Laboratory	40 Marks	60 Marks
3.	Laboratory Courses	50 Marks	50 Marks
4.	Project Work	40 Marks	60 Marks
5.	All other EEC Courses	100 Marks	----
6.	Value added Courses	100 Marks	----

There is no evaluation for Audit Courses. However, minimum attendance requirement as per clause 7 shall be satisfied.

Every teacher is required to maintain an 'ATTENDANCE AND ASSESSMENT RECORD' for every semester, which consists of attendance marked in each theory / Laboratory/EEC/AC class, the assessment marks and the record of class work (topics covered), separately for each course handled by the teacher.

This should be submitted to the Head of the Department periodically (at least three times in a semester) for checking the syllabus coverage and the records of assessment marks and attendance. The Head of the Department will affix his/her signature and date after due verification. At the end of the semester, the record should be verified by the Head of the Department who shall keep this document in safe custody (for five years).

The records of attendance and assessment of both current and previous semesters should be available for inspection.

7.1. Assessment for Theory Courses

For Theory Courses including mandatory courses out of 100 marks, the maximum marks for Continuous assessment is fixed as 40 and the End Semester Examination carries 60 marks. The End Semester Examination for theory courses including mandatory courses will be of 3 Hours duration and shall normally be conducted for a maximum of 100 marks between November /December during the Odd Semesters and between April/May during the Even Semesters.

End semester Examination is mandatory requirement for passing the Course. Continuous Assessment is based on the performance of the Students in tests and tutorial or objective type tests. Three assessments of equal weightage will be conducted by the Controller of Examinations. The total marks obtained in the assessments put together shall be reduced to 40 marks and rounded off to the nearest integer.

In case a student misses the assessment due to medical reasons (hospitalization / accident / specific illness) or due to participation in the College / University / State / National / International level Sports events with prior permission from the HOD, a Re-assessment may be given at the end of the semester after getting approval from the Head of the Department through the concerned Course handling faculty.

To arrive the Continuous Assessment Mark, the following guidelines are to be followed

S.No.	Category Details	Marks
1	Assessment I	15
2	Assessment II	15
3	Employability online Test/Mini Projects/ Scopus indexed publications/ Online MCQs/ Case studies/Tutorials/Innovative Projects idea generation/ Problem solving and Implementation	10
	Total	40

In case a student misses the assessment due to medical reasons (hospitalization / accident / specific illness) or due to participation in the College / University / State / National / International level Sports events with prior intimation from the Head of the institution, a Reassessment may be given after getting approval from the Head of the Department through the concerned Faculty advisor.

7.2. Assessment for Theory Courses with Laboratory Component

The maximum marks for Internal Assessment shall be 40 in case of theory courses with Laboratory component. (i.e.) Each course shall be evaluated for a maximum of 100 marks as shown below:

Sl. No	Category of Courses	Internal Assessment(40)		End-Semester Examination
		Assessment for Theory (T)	Assessment for Laboratory (L)	
1	Theory Courses with Laboratory Component	30 (T) (Two assessments)	10 (L) (One assessment)	60 (T)

7.3. Assessment for Laboratory Courses

Each laboratory Courses shall be evaluated for a maximum of 100 marks. Every laboratory exercise / experiment shall be evaluated based on the student performance during the laboratory class and the student's record shall be maintained. The End Semester Examination will be conducted for 3 hours

duration and shall normally be conducted for a maximum of 100 marks by Examiner other than the concerned laboratory course handling faculty.

Sl. No	Description	Marks
1	Internal Assessment Average mark awarded for Design and conduct of experiments(10 Marks), observation / coding / implementation (10 Marks),Results (10 Marks), Viva-Voce (10 Marks) and Record (10 Marks) in regular class works	50
2	End Semester Practical Examinations conducted by Internal Examiner appointed by Controller of Examinations	50
	Total	100

Examination shall normally be conducted between October and December during the odd semesters and between April and June during the even semesters.

7.4. Assessment for Project Work

Project work may be allotted to a single student or to a group of students not exceeding 4 per group. The Head of the department shall constitute a review committee for project work. There shall be three reviews during the semester by the review committee.

The student shall make presentation on the progress made by him / her before the committee. The marks obtained in the three reviews will be as indicated in clause 7.4.1.

The continuous assessment and End Semester Examinations marks for Project Work and the Viva-Voce Examination will be distributed as indicated below.

7.4.1 The project work shall be evaluated for a maximum of 100 marks, of which continuous Assessments carry 40 marks (total) and the End Semester Examination (Project report evaluation and Viva- Voce Examination) carries 60 marks. The project report shall be submitted as per the approved guidelines. The evaluation of the project reports will carry 20 marks and the same mark shall be awarded to every student within the project group for the project report. The viva-voce examination shall carry 40 marks and marks are awarded to each student of the project group based on the individual performance in the viva-voce examination.

Continuous Assessment 40 Marks (By supervisor and Committee Members)			End semester Examination 60 Marks			
Review I	Review II	Review III	Project Report Evaluation 20 marks		Viva-Voce 40 marks	
10	10	20	Internal	External	Internal	External
			10	10	20	20

7.4.2 If a student fails to submit the project report on or before the specified deadline, he/she is deemed to have failed in the Project Work and shall re-register for the same in a subsequent semester.

7.5. Assessment for Internship

The Internship shall carry 100 marks and shall be evaluated through continuous assessment only as per Table. At the end of the Industrial Training / Internship, the student shall submit a certificate from the organization where the student has undergone training and a brief report about the training. The evaluation will be made based on this report, presentation and a Viva Voce Examination conducted internally by a three member Departmental Committee consisting of one coordinator and two members constituted by the Head of the Department. Certificates (issued by the Organization) submitted by the student shall be attached to the mark list and sent to the Controller of Examinations by the Head of the Department.

INTERNSHIP (INTERNAL ASSESSMENT)		
REPORT	PRESENTATION	VIVA VOCE
40	30	30

8. GPA AND CGPA CALCULATION

After results are declared, Grade Sheets will be issued to each student which will contain the following details:

- The list of courses registered during the semester and the grades scored.
- The grade point average (GPA) for the semester
- The cumulative grade point average (CGPA) of all courses registered from first semester onwards.

During each semester, the list of courses registered and the grades scored in each course are used to compute the Grade Point Average (GPA). GPA is the ratio of the sum of the products of the number of credits of courses registered and the grade points corresponding to the grades scored in those courses, taken for all the courses, to the sum of the number of credits of all the courses in the semester.

$$\text{GPA / CGPA} = \frac{\sum_{i=1}^n C_i \text{GP}_i}{\sum_{i=1}^n C_i}$$

Where, C_i - is the Credits assigned to the course

GP_i - is the grade point corresponding to the letter grade obtained for each course

n - is number of all Courses successfully cleared during the particular semester in the case of GPA and during all the semesters in the case of CGPA.

CGPA will be calculated in a similar manner, considering all the courses enrolled from first semester. "RA" and "SA" grades will be excluded for calculating GPA and CGPA.

The credits earned through one / two credit courses shall not be considered for calculating GPA and CGPA.

If a student studies more number of electives (PEC/OEC) than required as per the student's programme curriculum, the courses with higher grades alone will be considered for calculation of CGPA.

9. ELIGIBILITY FOR THE AWARD OF DEGREE

A student shall be declared to be eligible for the award of the B.E. / B.Tech. Degree provided the student as

- i. Successfully gained the required number of total credits as specified in the curriculum corresponding to the student's programme within the stipulated time.
- ii. Successfully completed the course requirements, appeared for the End-Semester examinations and passed all the subjects prescribed in all the 8 semesters within a maximum period of 7 years reckoned from the commencement of the first semester to which the candidate was admitted.
- iii. Successfully passed any additional courses prescribed by the Board of Studies.
- iv. Successfully completed the NCC / NSS / NSO / YRC requirements.
- v. No disciplinary action is pending against the student.
- vi. The award of Degree must have been approved by the Syndicate of the University.